

LECTURE 11 · DISCRETE MATHEMATICS

Sets turn rules into collections

What belongs? What overlaps? What changes?



Three requests. How many permissions

An app requests camera, location, camera

$\{\text{camera}, \text{location}, \text{camera}\}$

$\{\text{camera}, \text{location}\}$: **two permissions**

A repeated element adds nothing

A set records membersh

An unordered collection of distinct objec

Belongs

$$a \in A$$

Does not belong

$$a \notin A$$

$$\{1, 2, 2, 3\} = \{3, 1, 2\}$$

Empty is different from containing em

No elements

$$\emptyset = \{\}, \quad |\emptyset| = 0$$

One element

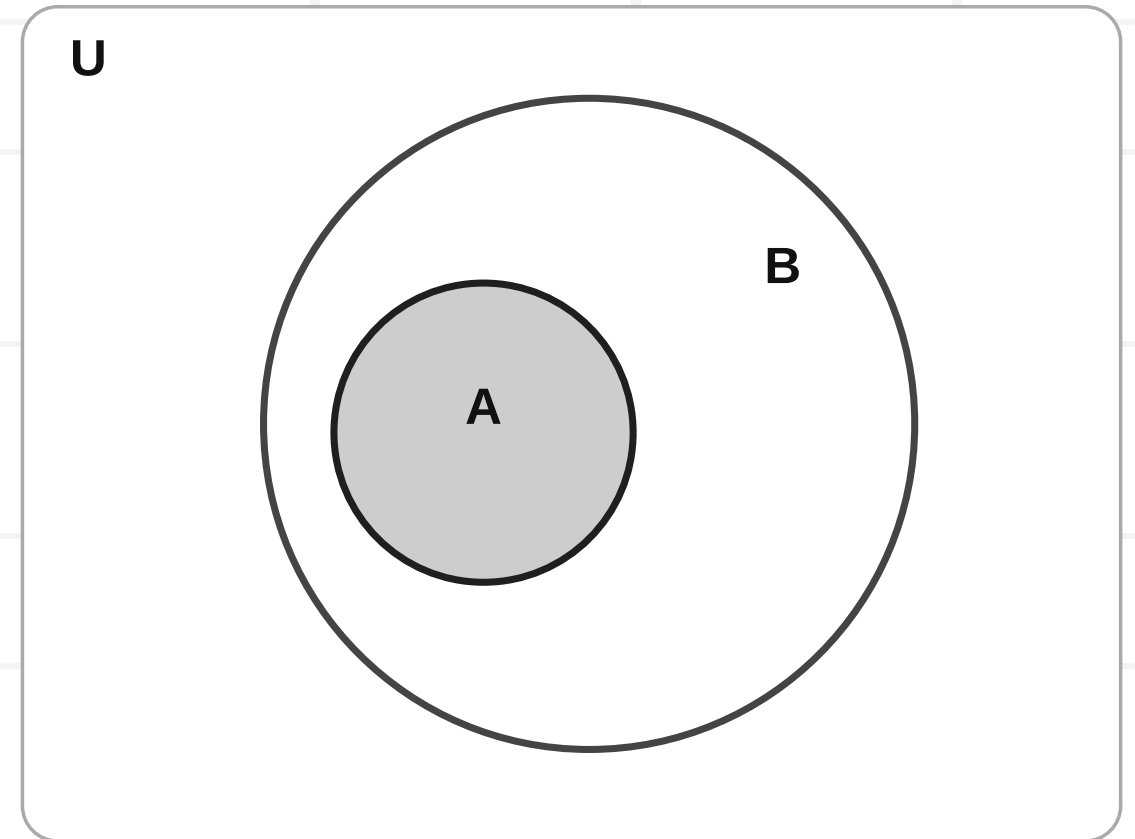
$$\{\emptyset\}, \quad |\{\emptyset\}| = 1$$

A singleton has exactly one element

A subset keeps every element

Every set has $\emptyset$ and itself as subsets. A proper subset also leaves something out

$$A \subseteq B \iff \forall x (x \in A \Rightarrow x \in B)$$



$$A \subsetneq B \iff A \subseteq B \text{ and } A \neq B$$

EXERCISE 1 · TRY FIRST

Is it listed? Or is every piece listed?

Let $T = \{0, \{0\}, \{1\}\}$. True or false?

01 $\{0\} \in T$

02 $\{0\} \subseteq T$

03 $\{1\} \subseteq T$

04 $\emptyset \in T$ and $\emptyset \subseteq T$

$X \in T$: is X itself listed? $X \subseteq T$: is each element of X listed?

The power set collects every sub

$$\mathcal{P}(S) = \{A : A \subseteq S\}. \text{ Let } S = \{0, 1\}.$$

$$\mathcal{P}(S) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$$

$$|S| = n \Rightarrow |\mathcal{P}(S)| = 2^n. \quad \emptyset \in \mathcal{P}(S), \quad S \in \mathcal{P}(S).$$

A power set can contain sets of sets

Let $R = \{1, \{1\}\}$.

01 Find $|R|$ and $|\mathcal{P}(R)|$.

02 List $\mathcal{P}(R)$.

03 Is $\{1\} \in \mathcal{P}(R)$? Is $\{\{1\}\} \in \mathcal{P}(R)$?

$$X \in \mathcal{P}(R) \iff X \subseteq R$$

Five operations. Five membership rules.

$$U = \{1, \dots, 6\}, \quad A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5\}.$$

Operation	x belongs when	Example
Union $A \cup B$	$x \in A \vee x \in B$	$\{1, 2, 3, 4, 5\}$
Intersection $A \cap B$	$x \in A \wedge x \in B$	$\{3, 4\}$
Difference $A \setminus B$	$x \in A \wedge x \notin B$	$\{1, 2\}$
Symmetric difference $A \triangle B$	x is in exactly one	$\{1, 2, 5\}$
Complement A^c	$x \in U \wedge x \notin A$	$\{5, 6\}$

Rosen writes $A \oplus B$ for $A \triangle B$ and $A - B$ for $A \setminus B$.

Compute each set. Then check with b

$$U = \{1, \dots, 8\}, \quad A = \{1, 2, 3, 6\}, \quad B = \{2, 4, 6, 8\}.$$

$$01 \quad A \cup B, \quad A \cap B, \quad B \setminus A, \quad (A \cup B)^c$$

$$02 \quad \text{Bit strings for } A \text{ and } B, \text{ in the order } 1, \dots, 8.$$

$$03 \quad \text{Bitwise AND and XOR. Decode bo}$$

Bit i is 1 exactly when element i belongs. OR, AND, XOR = union, intersection, symmetric difference

Prove set equality from both sides

Claim: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Translate to logic

$$x \in A \wedge \neg(x \in B \vee x \in C)$$

De Morgan: $x \in A, x \notin B, x \notin C$.

Left $\subseteq$ right

$$x \in A \setminus B \wedge x \in A \setminus C$$

Take any x on the left; the three facts put it in both differences

Right $\subseteq$ left

$$x \in A, x \notin B, x \notin C$$

So $x \notin B \cup C$, and $x \in A \setminus (B \cup C)$.

$$X = Y \iff X \subseteq Y \text{ and } Y \subseteq X$$

A membership table checks all eight cases

1 = x is in the set. 0 = x is not in the set

A	B	C	$B \cup C$	$A \setminus (B \cup C)$	$A \setminus B$	$A \setminus C$	$(A \setminus B) \cap (A \setminus C)$
1	1	1	1	0	0	0	0
1	1	0	1	0	0	1	0
1	0	1	1	0	1	0	0
1	0	0	0	1	1	1	1
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Columns 5 and 8 agree in every row, so the two sets are equal

Prove it for every set, or break it on

Decide. Then give a proof or one checked counterexample

01 If $A \setminus C = B \setminus C$, must $A = B$?

02 If $A \subseteq B$, must $\mathcal{P}(A) \subseteq \mathcal{P}(B)$?

03 If $A \triangle C = B \triangle C$, must $A = B$?

A false claim needs one specific example. A true claim needs an arbitrary example

With three sets, restore the triple over

Two sets: $|A \cup B| = |A| + |B| - |A \cap B|$. Three sets need one more correction

$$\begin{array}{|l} \text{Add the sets} \\ |A| + |B| + |C| \end{array}$$

$$\begin{array}{|l} \text{Subtract pairwise overlap} \\ -|A \cap B| - |A \cap C| - |B \cap C| \end{array}$$

$$\begin{array}{|l} \text{Add the triple overlap} \\ +|A \cap B \cap C| \end{array}$$

The result is $|A \cup B \cup C|$. A triple member counted $3 - 3 + 1 = 1$ time.

Count multiples with inclusion–exclusion

In $\{1, \dots, 100\}$, let D_m be the set of multiples of m .

$$|D_m| = \lfloor 100/m \rfloor, \quad D_2 \cap D_3 = D_6$$

01 How many are divisible by 2, 3, or 5?

02 How many are divisible by exactly one of the

Work outward from the centre: triple overlap, then pairs, then single

List the elements. Or give the ru

Roster

$$E = \{0, 2, 4, 6\}$$

Set builder

$$E = \{2k : k \in \mathbb{Z}, 0 \leq k \leq 3\}$$

The colon means “such that

The familiar number se

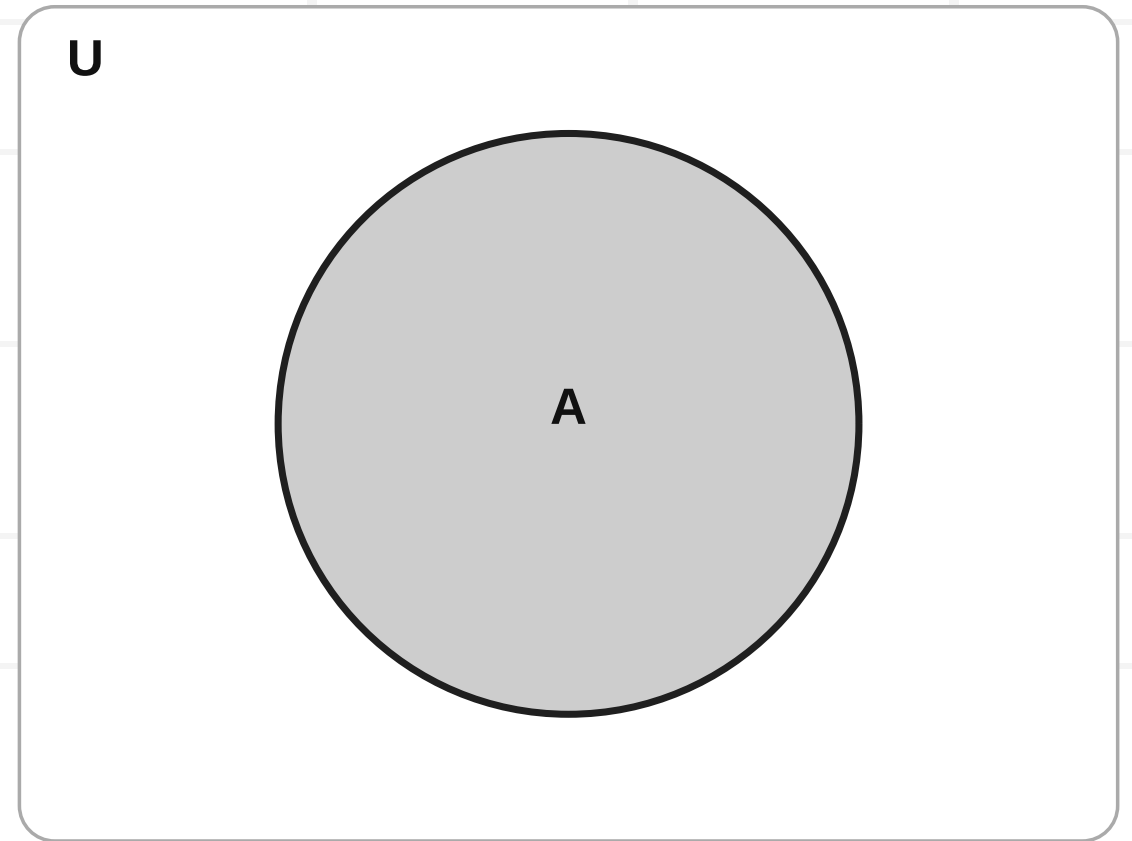
Set	Name	Members
$\mathbb{N}$	Natural number:	$0, 1, 2, \dots$
$\mathbb{Z}$	Integers	$\dots, -2, -1, 0, 1, 2, \dots$
$\mathbb{Z}^+$	Positive integer:	$1, 2, 3, \dots$
$\mathbb{Q}$	Rational number	$p/q : p, q \in \mathbb{Z}, q \neq 0$
$\mathbb{R}; \mathbb{R}^+$	Reals; positive real	$\mathbb{R}^+ = \{x \in \mathbb{R} : x > 0\}$
$\mathbb{C}$	Complex number	$a + bi : a, b \in \mathbb{R}, i^2 = -1$

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$$

Choose the universe first

U contains every object currently under discussion

$$A \subseteq U$$



Vowels: letters. App permissions: available permissions

One rewrite proves the inclusion

$$A = \{6r + 12 : r \in \mathbb{Z}\}, \quad B = \{3s : s \in \mathbb{Z}\}.$$

$$A \subseteq B \quad \text{but} \quad B \not\subseteq A$$

$$x = 6r + 12 = 3(2r + 4) \Rightarrow x \in B.$$

$3 \in B$, but $3 \notin A$: every element of A is divisible by 6

An arbitrary element proves inclusion. One counterexample refutes

Different rules. The same se

$$A = \{2r : r \in \mathbb{Z}\}, \quad B = \{2s - 2 : s \in \mathbb{Z}\}.$$

$$A = B \iff A \subseteq B \wedge B \subseteq A$$

Each new element doubles the choic

Add $x \notin S$. Each old subset A gives two choices

$$|S| = n \implies |\mathcal{P}(S)| = 2^n$$

Leave it out

A

Put it in

$A \cup \{x\}$

Include or exclude each element

How many permission bundles

Available permissions: camera, location, microphone. Any subset is allowed

01 **How many bundles**

02 **How many contain camera**

ACT II · SET OPERATION

Combine membership rule

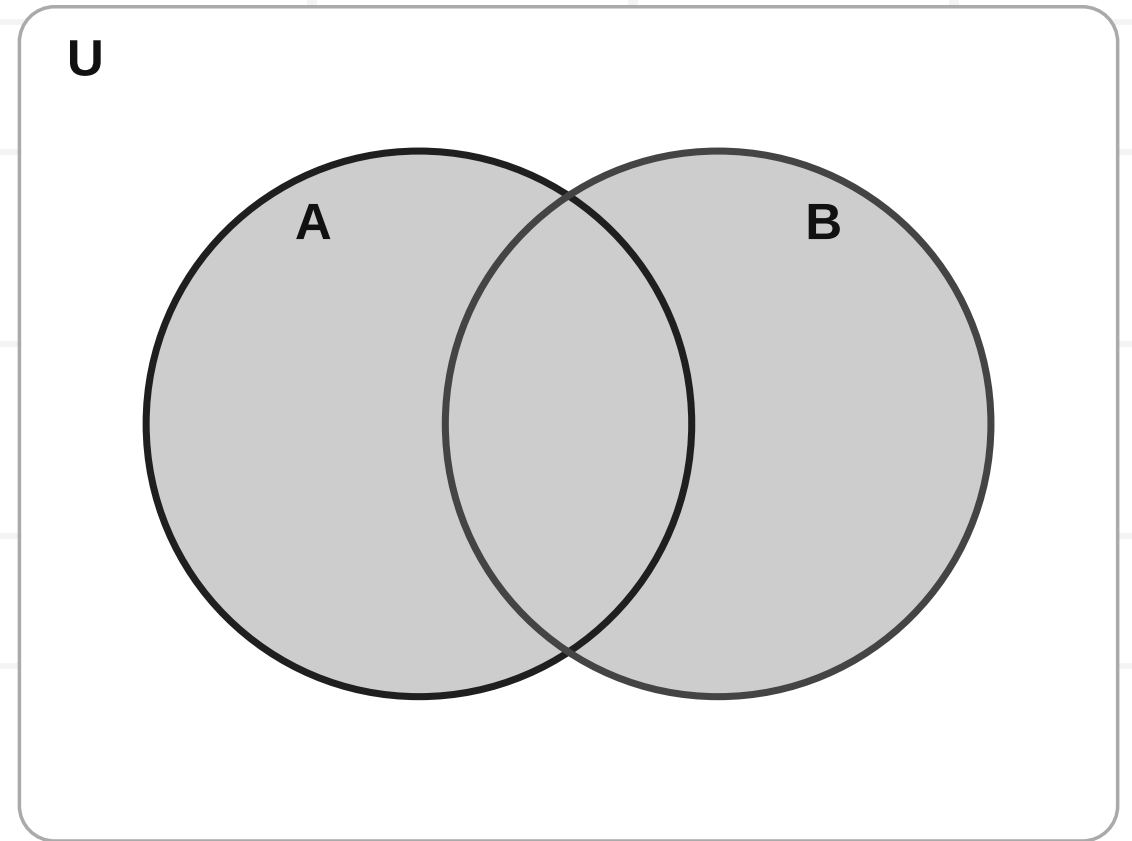
OR. AND. NOT. Now applied to collectio



Union means at least or

Photos tagged “night” OR “city”

$$x \in A \cup B \iff x \in A \vee x \in B$$

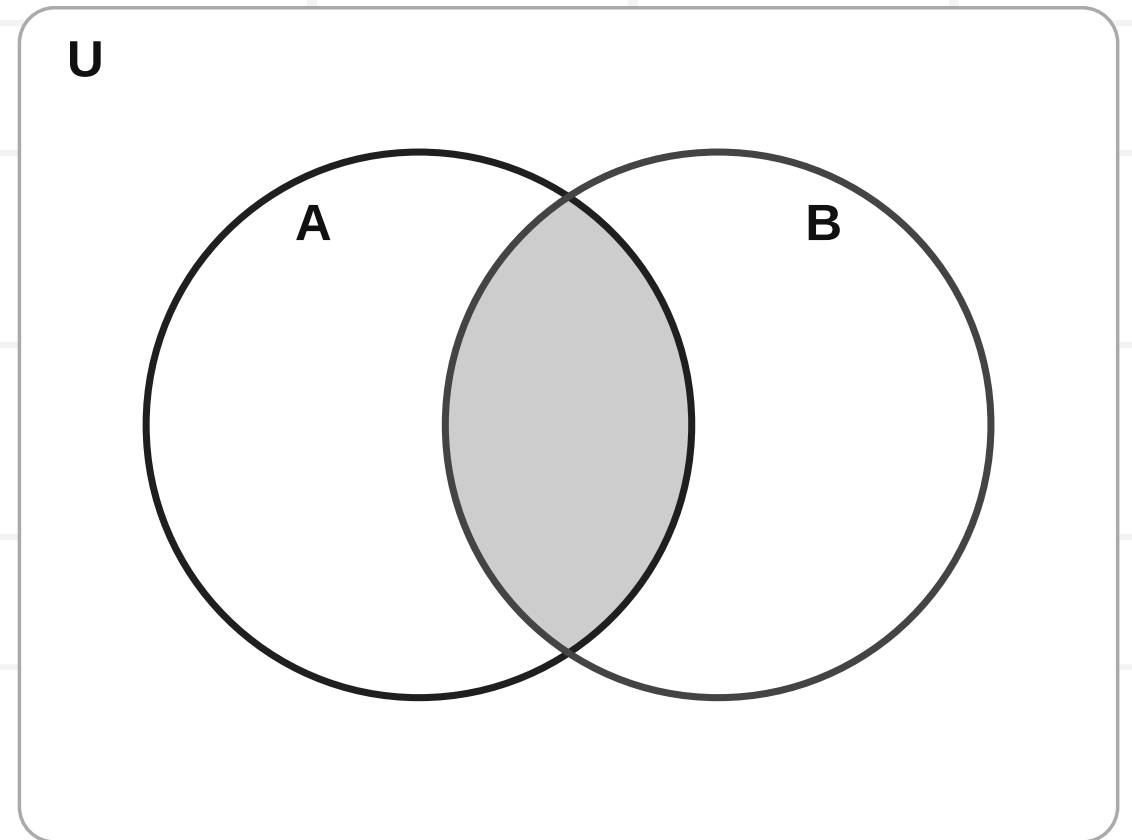


The overlap is included

Intersection means both

Photos tagged “night” AND “city”

$$x \in A \cap B \iff x \in A \wedge x \in B$$

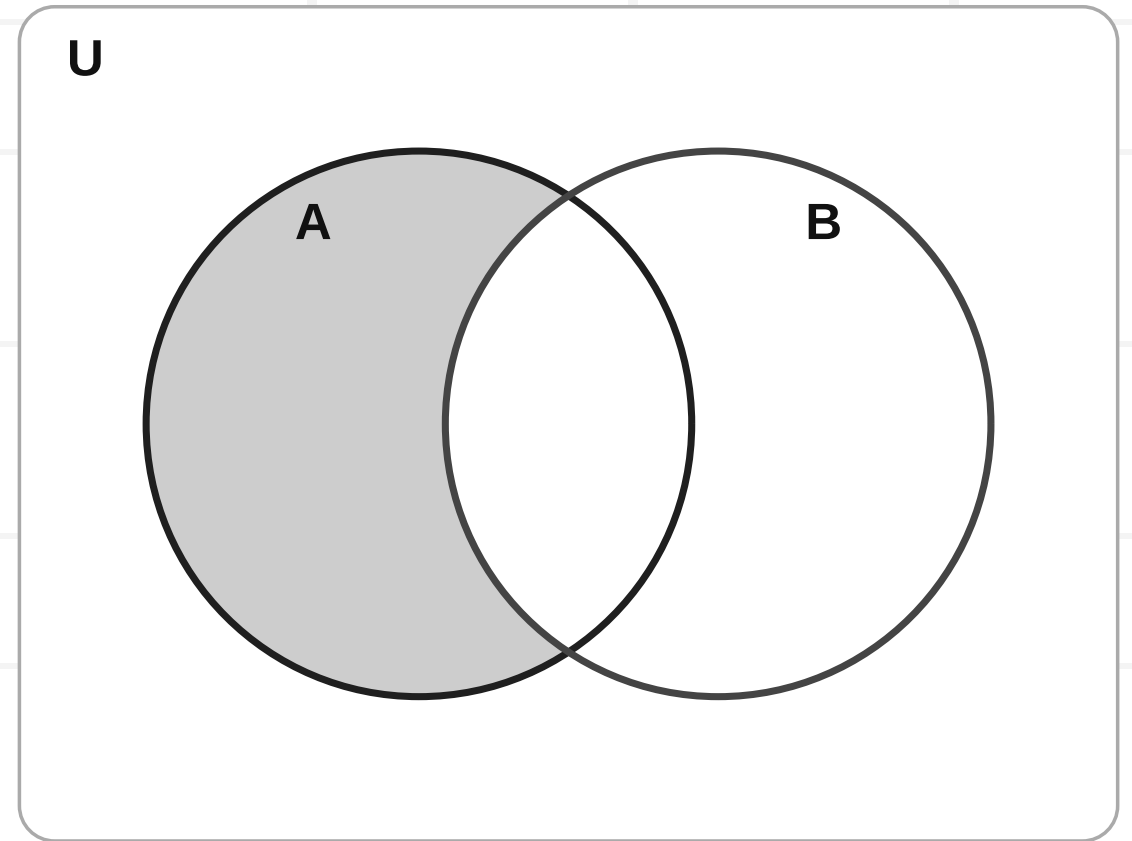


Only the overlap survives

Difference removes the second s

$$A = \{1, 3, 5\}, B = \{1, 2, 3\}$$

$$A \setminus B = \{x \in A : x \notin B\}$$

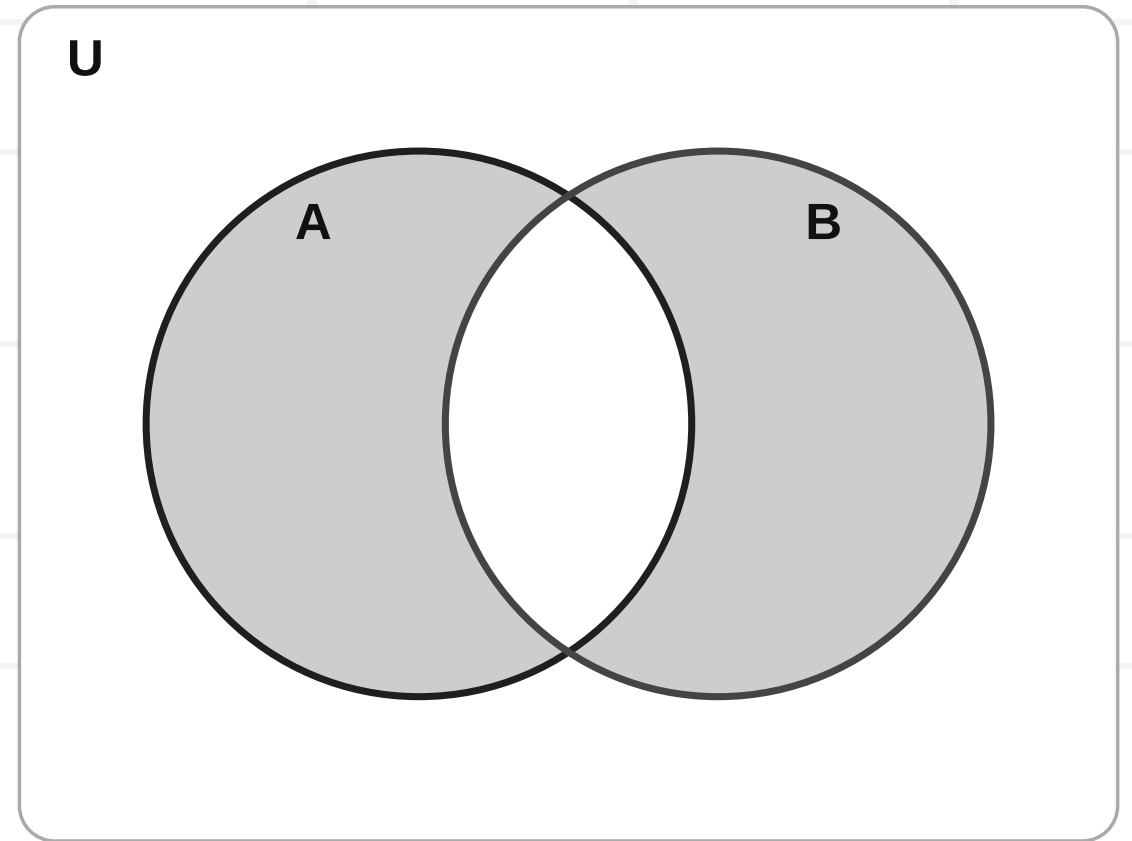


$$A \setminus B = \{5\}, \quad B \setminus A = \{2\}$$

Symmetric difference means exactly c

One tag, but not both

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

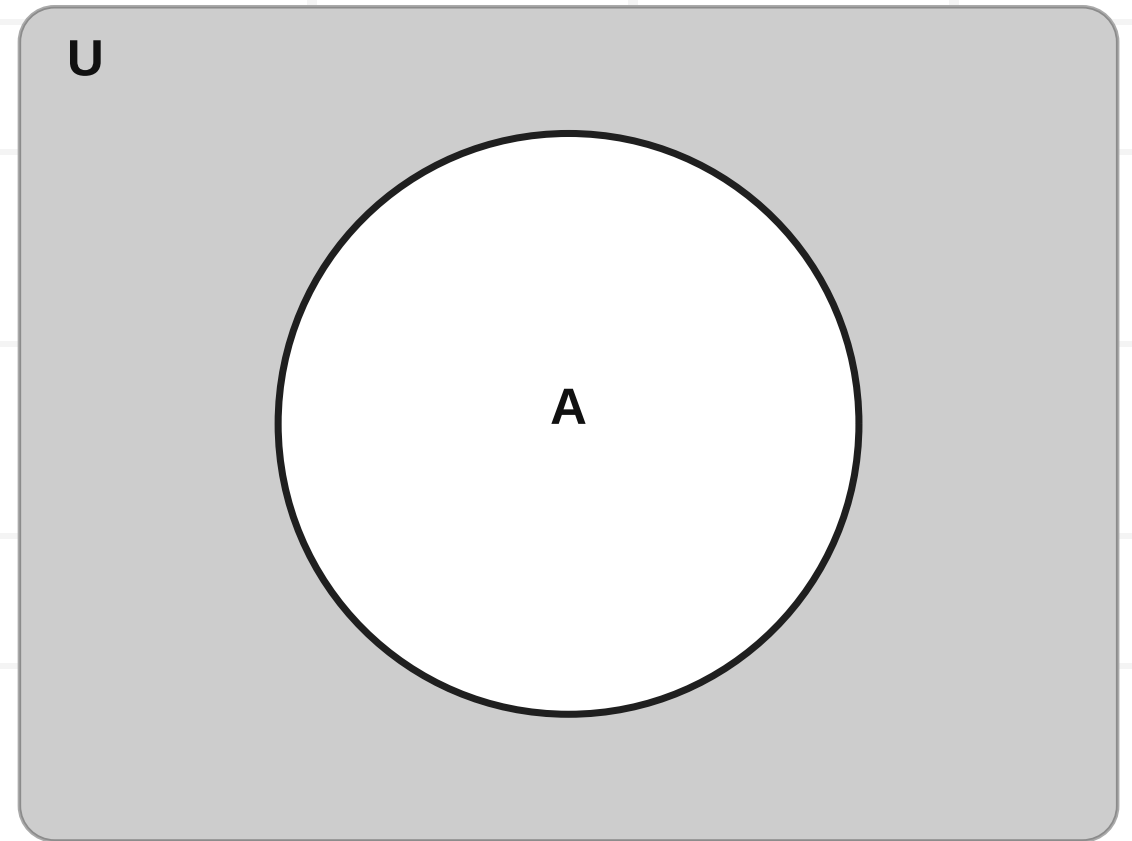


$$A \triangle A = \emptyset, \quad A \triangle A \triangle A = A$$

Complement depends on the univer

$$U = \{1, 2, 3, 4\}, \quad A = \{1, 3\}.$$

$$A^c = U \setminus A$$

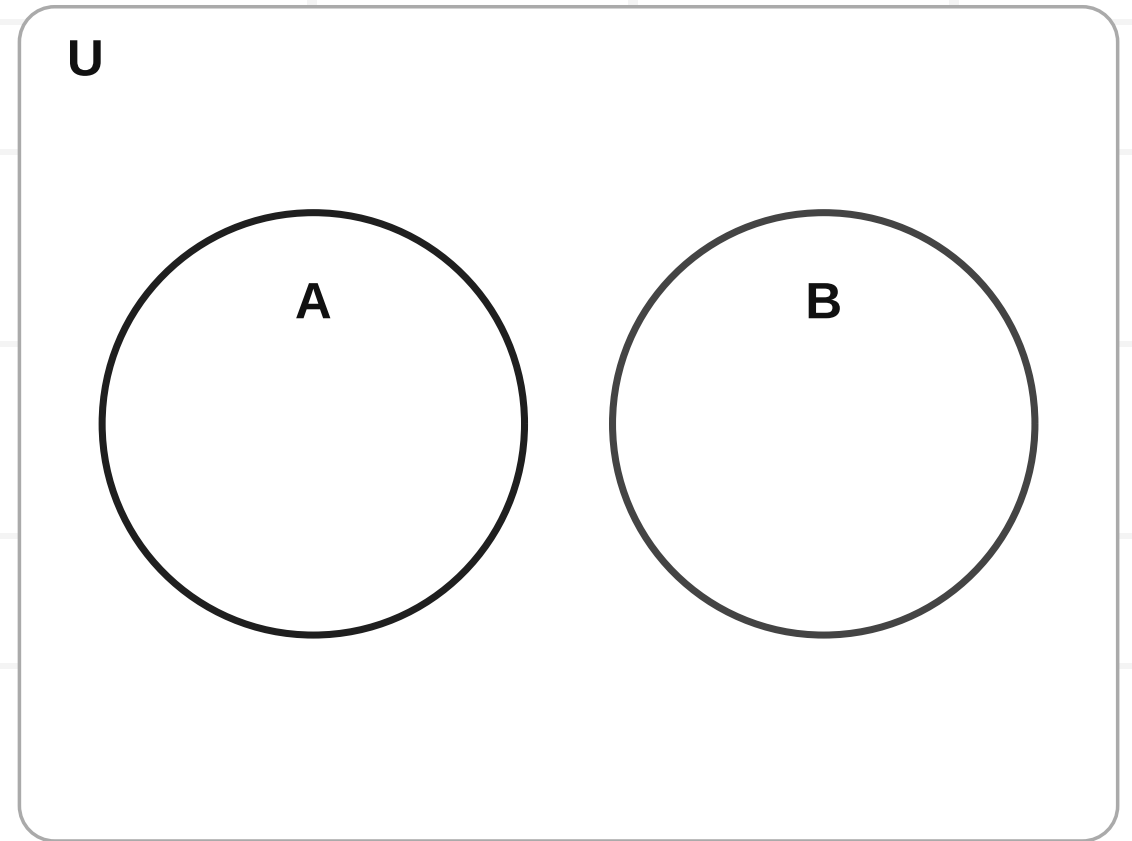


$A^c = \{2, 4\}$. Change U ; the answer may change

Disjoint means no shared element

Even IDs and odd IDs never overlap

$$A \cap B = \emptyset$$



$\emptyset$ is disjoint from every set, including itself

Prove a set law one element at a time

Fix an arbitrary element x in I

Complement of intersection

$$x \in (A \cap B)^c \iff \neg(x \in A \wedge x \in B)$$

Use De Morgan

$$\iff x \notin A \vee x \notin B$$

Translate back

$$\iff x \in A^c \cup B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Set laws mirror logi

Law	Union	Intersection
Idempotent	$A \cup A = A$	$A \cap A = A$
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Identity	$A \cup \emptyset = A$	$A \cap U = A$
Dominator	$A \cup U = U$	$A \cap \emptyset = \emptyset$

Reference: read $\cup$ as OR and $\cap$ as AND

Negation swaps union and intersect

Law	Identity
De Morgar	$(A \cup B)^c = A^c \cap B^c ; (A \cap B)^c = A^c \cup B^c$
Complemen	$A \cup A^c = U ; A \cap A^c = \emptyset$
Double complemer	$(A^c)^c = A ; U^c = \emptyset ; \emptyset^c = U$
Absorptior	$A \cup (A \cap B) = A ; A \cap (A \cup B) = A$

Distribute either operation over the other

AND over OR

$$A \cap (B \cup C) \\ = (A \cap B) \cup (A \cap C)$$

OR over AND

$$A \cup (B \cap C) \\ = (A \cup B) \cap (A \cup C)$$

Membership gives the familiar distributive laws of logic

Watch the endpoint

$$U = \mathbb{R}, \quad A = (-1, 0], \quad B = [0, 1).$$

$$01 \quad A \cup B \quad ; \quad A \cap B$$

$$02 \quad B \setminus A \quad ; \quad A^c$$

() excludes an endpoint. [] includes

“Some” means union. “Every” means intersect

At least one index

$$x \in \bigcup_{i=1}^n A_i \iff \exists i \in \{1, \dots, n\} : x \in A_i$$

Every index

$$x \in \bigcap_{i=1}^n A_i \iff \forall i \in \{1, \dots, n\} : x \in A_i$$

For infinitely many sets, use all positive integer indices

Shrinking intervals still share one point

$$A_i = (-1/i, 1/i), \quad i = 1, 2, 3, \dots$$

$$01 \quad A_1 \cup A_2 \cup A_3 \quad ; \quad A_1 \cap A_2 \cap A_3$$

$$02 \quad \bigcup_{i=1}^{\infty} A_i \quad ; \quad \bigcap_{i=1}^{\infty} A_i$$

Any nonzero x eventually falls outside